

RAG Retrieval Evaluation Metrics: Definitions, Mathematics, and Worked Examples

Companion Reference for the Agentic BI Platform – Phase 5 Evaluation Harness

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1 Introduction and scope

The Phase 5 evaluation harness (`run_retrieval_eval.py`) was extended from a simple pass/fail check into six standard information-retrieval metrics: **Hit Rate@K**, **Recall@K**, **Precision@K**, **Full Coverage** (the original pass/fail criterion, kept alongside the new metrics), **Mean Reciprocal Rank (MRR)**, and **NDCG@K**. This document explains each one precisely – the formula, what question it answers, and a worked numerical example – so every number the harness prints can be reproduced by hand and explained without hand-waving.

1.1 The one caveat that applies to every metric in this document

All six metrics below are computed at **source level** (which document a chunk came from), not at chunk level or claim level. A test case is labeled with a set of **expected_sources** – the documents that should appear among the retrieved results – not with which exact 250-character chunk should appear, and not with which specific sentence of the final generated answer should be traceable to which chunk. This is a real simplification, made deliberately for this project’s retrieval-quality harness, and it matters for what these numbers can and cannot tell you:

- They can tell you whether the retriever surfaced the *right document*, and how high it ranked.
- They cannot tell you whether the *specific chunk* returned actually contained the fact needed to answer the question (this project caught several cases, documented separately in the decision log, where a source-level “pass” was hiding a chunk that didn’t actually contain the relevant sentence).
- They cannot tell you whether the final LLM-generated answer was correct (that requires chain-level, answer-grounded evaluation – the subject of a later project phase, not this one).

Keep this in mind throughout: a high score on every metric in this document is necessary, but not sufficient, evidence that the system produces correct answers.

1.2 Shared setup

Every metric operates on the same two inputs for a given test case:

- E : the set of **expected sources** (ground truth) – e.g. $E = \{\text{kpi_definitions.md}\}$ for a single-source question, or $E = \{\text{customer_segmentation.md}, \text{kpi_definitions.md}\}$ for a cross-document question.
- $\hat{S} = (\hat{s}_1, \hat{s}_2, \dots, \hat{s}_m)$: the **ordered list of unique sources** actually retrieved, in rank order, after collapsing repeated chunks from the same source down to their first (best) rank. If a retriever returns chunks from `refund_and_returns_policy.md` at positions 1, 2, and 4, that source appears only once in $\hat{S}$, at rank 1.

Given a cutoff k (the retrieval depth being evaluated, e.g. $k = 4$ or $k = 2$), define the **top- k retrieved set**:

$$\hat{S}_{@k} = (\hat{s}_1, \dots, \hat{s}_{\min(k, m)})$$

Every metric below is defined per test case first, then **averaged across all test cases with at least one expected source** (negative controls, which have no expected source, are excluded from every average – they are inspected manually instead, since a retriever must always return *something*, so no numeric score can distinguish “correctly found nothing relevant” from “returned garbage”).

2 Hit Rate@K

2.1 Question answered

“Did we retrieve at least one useful document?” This is the most forgiving metric in the set – it only asks whether any part of the right answer showed up, not whether all of it did.

2.2 Formula

$$\text{HitRate@K} = \mathbb{I} \left[E \cap \hat{S}_{@k} \neq \emptyset \right]$$

where $\mathbb{I}[\cdot]$ is the indicator function (1 if the condition is true, 0 otherwise). Averaged across N test cases:

$$\text{HitRate@K} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left[E_i \cap \hat{S}_{@k,i} \neq \emptyset \right]$$

2.3 Worked example

Case 4 in this project’s eval set: *“Is a customer still enterprise if they haven’t ordered in a while?”*, with $E = \{\text{customer_segmentation.md}, \text{kpi_definitions.md}\}$. Suppose at $k = 2$, **Basic** retrieval returns:

$$\hat{S}_{@2} = (\text{customer_segmentation.md}, \text{customer_segmentation.md})$$

(collapsed to one unique entry, since both chunks share a source). The intersection $E \cap \hat{S}_{@2} = \{\text{customer_segmentation.md}\}$, which is non-empty, so **HitRate@2 = 1** for this case – even though a full second source (**kpi_definitions.md**) is entirely missing. This is the precise reason Hit Rate is not, by itself, a trustworthy signal for multi-source questions: it would report “success” on a case that this project’s stricter Full Coverage metric (below) correctly flags as a failure.

3 Recall@K

3.1 Question answered

“Of everything that should have been retrieved, what fraction actually was?”

3.2 Formula

$$\text{Recall@K} = \frac{|E \cap \hat{S}_{@k}|}{|E|}$$

Note the denominator is the size of the **ground truth** set, not the size of the retrieved set – this is what distinguishes Recall from Precision (below).

3.3 Worked example

Same case 4, same retrieval at $k = 2$: $|E| = 2$, $|E \cap \hat{S}_{@2}| = 1$ (only `customer_segmentation.md` was retrieved).

$$\text{Recall}@2 = \frac{1}{2} = 0.5$$

Compare this to Hit Rate’s binary 1 for the identical retrieval: Recall correctly communicates that only *half* of the needed evidence came back, where Hit Rate’s binary framing obscured that. This is why this project’s harness reports both – they answer genuinely different questions from the same underlying data, and printing only one would hide the other’s information.

3.4 A note on the ceiling of Recall@K

If $|E| > k$ (more distinct sources are needed than the retrieval depth allows), Recall@K can never reach 1.0, no matter how good the retriever is. For example, at $k = 1$ with $|E| = 2$, the best possible Recall@1 is 0.5. This is a real, structural ceiling worth remembering when reading Recall@K numbers at small k – a value below 1.0 does not automatically indicate poor retrieval quality; it may simply reflect that the cutoff is narrower than the number of documents the question genuinely requires.

4 Precision@K

4.1 Question answered

“Of everything we retrieved, what fraction was actually useful?” – the inverse framing of Recall.

4.2 Formula, as implemented in this project’s harness

$$\text{Precision}@K = \frac{|E \cap \hat{S}_{@k}|}{|\hat{S}_{@k}|}$$

Implementation note, worth being precise about: the denominator here is $|\hat{S}_{@k}|$ – the number of unique sources *actually retrieved* within the top- k window, which can be smaller than k itself if the retriever returned fewer than k distinct sources (this happens, for example, when a small- k BM25 retriever returns only 1 or 2 unique sources total). Some textbook definitions of Precision@K instead always divide by the nominal k , treating any “missing” slot as a non-relevant result by convention. This project’s implementation uses the former convention – precision among what was actually handed back, not among a hypothetical fixed-size slate. Both are legitimate; they simply answer subtly different questions, and it’s worth knowing which one you’re reading.

4.3 Worked example

Suppose, for the query *“What is the restocking fee percentage?”* (case 1, $E = \{\text{refund_and_returns_policy.md}\}$), a retriever at $k = 4$ returns:

$$\hat{S}_{@4} = (\text{refund_and_returns_policy.md}, \text{kpi_definitions.md})$$

(two unique sources, after collapsing duplicate chunks). Then $|E \cap \hat{S}_{@4}| = 1$, and $|\hat{S}_{@4}| = 2$:

$$\text{Precision@4} = \frac{1}{2} = 0.5$$

Exactly half of the distinct sources retrieved were actually relevant to this single-source question – the other half (`kpi_definitions.md`) was retrieved but not needed here.

5 Full Coverage (case-level pass/fail)

5.1 Question answered

“Did we get every single source this question needs – yes or no?” This is the original, stricter criterion this project’s harness used before the numeric metrics were added, and it remains valuable specifically because it does **not** average away partial success the way Recall@K does.

5.2 Formula

$$\text{FullCoverage} = \mathbb{K} \left[E \subseteq \hat{S}_{@k} \right]$$

Averaged across cases, this becomes the fraction of test cases that *fully* passed – the “X/9 cases fully retrieved” line printed at the bottom of every report throughout this project’s Phase 5 evaluation.

5.3 Why this is not redundant with Recall@K

Recall@K rewards partial credit (0.5 for getting half of a two-source requirement); Full Coverage does not – it is binary, all-or-nothing, per case. Both are useful, for different purposes: Recall@K is the right metric for tracking gradual improvement across many experiments (a technique that moves average Recall@K from 0.85 to 0.90 made real progress, even if it didn’t flip any single case from fail to pass); Full Coverage is the right metric for the specific, practical question “is this particular business question answerable end-to-end right now,” since a downstream RAG chain either has all the context it needs to answer correctly, or it doesn’t – there is no such thing as an LLM answer that is “50% grounded.”

6 Mean Reciprocal Rank (MRR)

6.1 Question answered

“On average, how far down the list do we have to look before we find the first relevant source?”

6.2 Formula

For a single case, the reciprocal rank is:

$$\text{RR} = \frac{1}{\text{rank of the first relevant source in } \hat{S}}$$

or 0 if no relevant source appears at all. Averaged across N cases:

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i}$$

Implementation note – an important one. As implemented in this project’s current harness, the search for “the first relevant source” runs over the **entire** retrieved list $\hat{S}$, not the truncated $\hat{S}_{@k}$ used by every other metric in this document. This means MRR, as currently computed, is **not** “MRR@K” – it can award partial credit for a relevant source found beyond the cutoff k , even in a report where every other metric (Recall@K, Precision@K, Hit Rate@K, NDCG@K) is strictly windowed to the same k . This is worth fixing for internal consistency (restricting the search to $\hat{S}_{@k}$ would make it genuinely “MRR@K,” matching the rest of the report), or at minimum documenting clearly, since a reader comparing all six numbers side by side would otherwise reasonably assume they all share the same retrieval window.

6.3 Worked example

For the query “*What counts as an active customer?*” (case 3, $E = \{\text{kpi_definitions.md}\}$), suppose the full retrieved list (unique sources, in rank order) is:

$$\hat{S} = (\text{customer_segmentation.md}, \text{kpi_definitions.md})$$

The first relevant source (`kpi_definitions.md`) is at rank 2:

$$\text{RR} = \frac{1}{2} = 0.5$$

If this were the only case in the evaluation set, $\text{MRR} = 0.5$. Averaged across, for example, three cases with reciprocal ranks 1.0, 0.5, 0.333 (relevant source found at ranks 1, 2, and 3 respectively):

$$\text{MRR} = \frac{1.0 + 0.5 + 0.333}{3} = \frac{1.833}{3} \approx 0.611$$

MRR is especially sensitive to *how far down* you have to look, which makes it a useful complement to Hit Rate@K: two retrievers can have identical Hit Rate@K (both eventually found the answer within the window) while having very different MRR (one found it at rank 1, the other at rank 4) – a real difference in practical usefulness that Hit Rate alone cannot distinguish, since Hit Rate only asks “yes or no,” never “how quickly.”

7 Normalized Discounted Cumulative Gain (NDCG@K)

7.1 Why this is the most information-rich metric in the set

Every metric above either ignores rank entirely (Recall@K, Precision@K, Full Coverage just ask “is it in the window, yes or no”) or only looks at the *first* relevant result (MRR). NDCG is the only metric here that accounts for **the rank of every relevant result**, not just whether one exists

or where the first one sits – which matters directly for a multi-source question like case 4, where getting both relevant sources at ranks 1 and 2 is genuinely better than getting them at ranks 1 and 4, even though every other metric above (except MRR, which only sees the first one) would treat those two outcomes identically.

7.2 Step 1: Discounted Cumulative Gain (DCG)

For binary relevance (a source is either relevant or it isn’t – this project’s ground truth has no notion of “partially relevant”), DCG at cutoff k is:

$$\text{DCG@K} = \sum_{j=1}^k \frac{\text{rel}_j}{\log_2(j+1)}$$

where $\text{rel}_j \in \{0, 1\}$ is 1 if the source at rank j is in E , else 0. The $\log_2(j+1)$ term in the denominator is the **discount**: a relevant result at rank 1 contributes $1/\log_2(2) = 1/1 = 1.0$ to the sum; the identical relevant result at rank 4 contributes only $1/\log_2(5) \approx 0.431$. The discount formalizes the intuition that a relevant result buried lower in the list is worth less to a user (or a downstream LLM prompt, which typically weighs earlier context more heavily) than the same result at the top.

7.3 Step 2: Ideal DCG (IDCG) – the normalizing denominator

Raw DCG is not comparable across cases with different numbers of expected sources – a case with 1 expected source has a different maximum possible DCG than a case with 2. IDCG is the DCG of the **best possible ordering**: every relevant source placed at the very top of the ranking, back to back:

$$\text{IDCG@K} = \sum_{j=1}^{\min(|E|, k)} \frac{1}{\log_2(j+1)}$$

7.4 Step 3: Normalize

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$

This ratio is always in $[0, 1]$: 1.0 means the retriever achieved the best possible ranking for that case (every relevant source at the very top of the window); 0 means nothing relevant appeared at all within the window.

7.5 Worked example, using this project’s actual case 4 (two expected sources)

Recall the retrieval-level finding, confirmed in this project’s Hybrid retrieval evaluation: at $k = 4$, Basic similarity search returned `customer_segmentation.md` at rank 1 and `kpi_definitions.md` at rank 4.

DCG@4:

$$\text{DCG@4} = \underbrace{\frac{1}{\log_2(2)}}_{\text{rank 1: relevant}} + \underbrace{\frac{1}{\log_2(5)}}_{\text{rank 4: relevant}} = 1.0 + 0.4307 = 1.4307$$

(Ranks 2 and 3, both `customer_segmentation.md` chunks – already counted at rank 1 – contribute nothing further once collapsed to unique sources.)

IDCG@4 (best possible: both relevant sources at ranks 1 and 2):

$$\text{IDCG@4} = \frac{1}{\log_2(2)} + \frac{1}{\log_2(3)} = 1.0 + 0.6309 = 1.6309$$

NDCG@4:

$$\text{NDCG@4} = \frac{1.4307}{1.6309} \approx 0.877$$

Compare this to the Hybrid-retrieval result for the same case, where RRF fusion placed `kpi_definitions.md` at rank 2 instead of rank 4:

$$\begin{aligned} \text{DCG@4}_{\text{Hybrid}} &= \frac{1}{\log_2(2)} + \frac{1}{\log_2(3)} = 1.0 + 0.6309 = 1.6309 \\ \text{NDCG@4}_{\text{Hybrid}} &= \frac{1.6309}{1.6309} = 1.0 \end{aligned}$$

This is the precise, numerical way to state what this project’s decision log described qualitatively: both **Basic** and **Hybrid** achieved the same **Full Coverage** (both eventually included `kpi_definitions.md` within $k = 4$) and the same **Recall@4** (1.0, since both sources appeared somewhere in the top 4) – but **NDCG@4** correctly distinguishes them: 0.877 for **Basic** (the second relevant source arrived late, at rank 4) versus a perfect 1.0 for **Hybrid** (both relevant sources arrived as early as they possibly could). **NDCG** is the one metric in this set that would have caught this quality difference even if both retrievers had passed Full Coverage and tied on every other metric.

8 Summary table

Metric	Answers	Range	Rank-aware?	Penalizes partial multi-source retrieval?
Hit Rate@K	Did we get <i>anything</i> useful?	$\{0, 1\}$ per case	No	No – one hit is enough
Recall@K	What fraction of what’s needed came back?	$[0, 1]$	No	Partially (proportional credit)
Precision@K	What fraction of what we retrieved was useful?	$[0, 1]$	No	N/A (measures noise, not completeness)
Full Coverage	Did we get <i>everything</i> needed?	$\{0, 1\}$ per case	No	Yes – all-or-nothing

Metric	Answers	Range	Rank-aware?	Penalizes partial multi-source retrieval?
MRR	How far to the <i>first</i> relevant hit?	$[0, 1]$	Yes (first hit only)	No – ignores everything after the first hit
NDCG@K	How well-ranked is <i>every</i> relevant hit?	$[0, 1]$	Yes (all hits)	Yes – rewards early placement of every source

Practical reading order, given this project’s actual findings: Full Coverage is the right first number to check (does this question work at all, right now); Recall@K and Hit Rate@K explain *why* a Full Coverage failure happened (nothing came back at all, versus some-but-not-all); MRR and NDCG@K are the right numbers for comparing two retrievers that *both* pass Full Coverage, since they are the only two metrics that can detect a genuine quality difference (as with **Basic** vs. **Hybrid** on case 4) that a binary pass/fail criterion would otherwise call a tie.

9 Appendix: worked mini-report for one hypothetical case

To see all six metrics computed side by side from the same raw data, consider a single hypothetical case with $E = \{A, B\}$ (two expected sources) at $k = 4$, where a retriever returns unique sources in this order: $\hat{S} = (A, C, D, B)$ (A at rank 1, an irrelevant C at rank 2, an irrelevant D at rank 3, B at rank 4).

- **Hit Rate@4** = 1 (at least one of $\{A, B\}$ appears)
- **Recall@4** = $\frac{|\{A, B\} \cap \{A, C, D, B\}|}{2} = \frac{2}{2} = 1.0$
- **Precision@4** = $\frac{2}{4} = 0.5$ (2 of the 4 retrieved sources were relevant)
- **Full Coverage** = 1 (both A and B present within the window)
- **MRR (unrestricted, as currently implemented)** = $\frac{1}{1} = 1.0$ (first relevant hit, A , is at rank 1)
- **NDCG@4:** $\text{DCG@4} = \frac{1}{\log_2 2} + \frac{1}{\log_2 5} = 1.0 + 0.431 = 1.431$; $\text{IDCG@4} = \frac{1}{\log_2 2} + \frac{1}{\log_2 3} = 1.0 + 0.631 = 1.631$; $\text{NDCG@4} = \frac{1.431}{1.631} \approx 0.877$

Notice that **Recall@4** and **Full Coverage** both report a perfect score, and **MRR** also reports a perfect score (since it only looks at the first hit, and the first hit happened to be perfect) – but **NDCG@4** correctly reports 0.877, **not** 1.0, because the second relevant source (B) arrived later than it ideally could have. This is the clearest possible illustration of why this project’s harness reports all six numbers rather than any single one: each metric is blind to a different kind of imperfection, and only by reading them together does the full picture of a retriever’s behavior become visible.

10 Appendix B: reading a real run – Basic similarity search at $k=2$

The hypothetical example above is useful for seeing the mechanics in isolation. This appendix does the same walkthrough against an actual printed report from this project’s harness, so every number below is something you can reproduce by rerunning the code, not a constructed illustration.

10.1 The aggregate report being explained

```
SOURCE-LEVEL AGGREGATE METRICS  --  Basic (k=2)
Recall@2:      0.944
Precision@2:   0.889
Hit Rate@2:    1.000
MRR@2:        0.944
NDCG@2:       0.916
Full Coverage: 0.889   (8/9 cases fully retrieved)
```

Nine of this project’s ten test cases have at least one expected source (the tenth, case 10, is the negative control and is excluded from every average, per the shared setup in the Introduction). Two of those nine cases – case 3 and case 4 – are imperfect on at least one metric, and walking through exactly *why* each number lands where it does is the fastest way to build real intuition for what these six numbers do and do not tell you.

10.2 Case 3: a single-source question with a near-miss at rank 2

“*What counts as an active customer?*”, with $E = \{\text{kpi_definitions.md}\}$ (one expected source). Suppose `Basic` at $k = 2$ returns the unique sources, in rank order:

$$\hat{S}_{@2} = (\text{customer_segmentation.md}, \text{kpi_definitions.md})$$

The correct source is present, but at rank 2, not rank 1 – the query’s casual phrasing (“what counts as”) pulled in `customer_segmentation.md` (which also discusses customer classification) slightly ahead of the document that actually defines the term.

- **Recall@2** = $\frac{|\{kpi\} \cap \hat{S}_{@2}|}{|\{kpi\}|} = \frac{1}{1} = 1.0$ – the one required source did eventually appear within the window.
- **Precision@2** = $\frac{1}{2} = 0.5$ – only one of the *two* unique sources retrieved was actually relevant; the other slot was “spent” on `customer_segmentation.md`.
- **Hit Rate@2** = 1 – at least one relevant source appeared.
- **RR@2** = $\frac{1}{2} = 0.5$ – the first relevant source was at rank 2, not rank 1.
- **NDCG@2**: $\text{DCG}@2 = \frac{1}{\log_2(3)} \approx 0.631$ (relevant source at rank 2); $\text{IDCG}@2 = \frac{1}{\log_2(2)} = 1.0$
(best case: relevant source at rank 1); $\text{NDCG}@2 = \frac{0.631}{1.0} \approx 0.63$.

Every metric here is telling a version of the same story – *found, but not at the top* – at a different level of detail. Recall and Hit Rate don’t notice the rank at all; Precision notices that a slot was

“wasted”; RR and NDCG both notice the rank itself, and NDCG expresses it as a clean fraction of the best-possible score rather than RR’s raw “1 divided by rank” framing.

10.3 Case 4: a two-source question where the second source never arrives

“Is a customer still enterprise if they haven’t ordered in a while?” – a two-source question:

$$E = \{ \text{customer_segmentation.md}, \text{kpi_definitions.md} \}$$

Basic at $k = 2$ returns two chunks, both from the same document:

$$\hat{S}_{@2} = (\text{customer_segmentation.md})$$

(a single unique source, since both retrieved chunks collapse to one entry). Notice that this single retrieved source happens to itself be one of the two expected sources – which produces a genuinely instructive pattern across the six metrics:

- **Recall@2** = $\frac{|\{seg, kpi\} \cap \{seg\}|}{2} = \frac{1}{2} = 0.5$ – only half of what’s needed came back.
- **Precision@2** = $\frac{1}{1} = 1.0$ – of the one unique source retrieved, all of it was relevant. A perfect precision score here is *not* good news on its own; it simply reflects that nothing irrelevant was retrieved, which is trivially true when only one thing was retrieved at all.
- **Hit Rate@2** = 1 – at least one relevant source (in fact, an expected one) is present.
- **RR@2** = $\frac{1}{1} = 1.0$ – the *first* relevant source is at rank 1. A perfect score.
- **NDCG@2**: $DCG@2 = \frac{1}{\log_2(2)} = 1.0$ (one relevant hit, at rank 1); $IDCG@2 = \frac{1}{\log_2(2)} + \frac{1}{\log_2(3)} \approx 1.631$ (best case: *both* relevant sources present, at ranks 1 and 2); $NDCG@2 = \frac{1.0}{1.631} \approx 0.613$.
- **Full Coverage** = 0 (FAIL) – `kpi_definitions.md` never appeared at all within the window.

This is the single most important pattern to internalize from this entire document: **Hit Rate@2 and RR@2 both report a perfect 1.0 for a case that Full Coverage correctly reports as a failure.** Neither metric is malfunctioning – both are answering the specific, narrower question they were designed to answer (“was *something* relevant found, and how fast”), and neither was ever designed to notice that a *second*, independently required piece of evidence is completely absent. Only Recall@2 (partial credit, 0.5), NDCG@2 (rank-aware partial credit, 0.613), and Full Coverage (strict all-or-nothing, a hard 0) surface the actual problem. This is exactly why this project’s harness prints all six numbers together rather than picking a single “headline” metric – reducing this case to any one number below Full Coverage would have hidden the real failure behind a misleadingly perfect score.

10.4 Reading the aggregate numbers with this in hand

Across all nine real cases, only case 3 and case 4 are imperfect on any metric; the remaining seven are perfect on all six. This is enough to explain every aggregate number reported:

$$\text{Recall@2 (avg)} = \frac{7 \times 1.0 + 0.5_{(\text{case 4})} + 1.0_{(\text{case 3})}}{9} = \frac{8.5}{9} \approx 0.944$$

$$\text{Full Coverage} = \frac{8}{9} \approx 0.889$$

(case 4 is the sole failure; case 3, despite an imperfect rank, still achieved *full* coverage since its one required source did appear somewhere within the window). The same pattern of “seven perfect cases, two cases each imperfect on a different subset of metrics” explains Precision@2 (0.889), MRR@2 (0.944 – note this average is *higher* than case 3’s own RR of 0.5 would suggest, precisely because case 4 contributes a perfect 1.0 to this average, since RR does not penalize a missing *second* source at all), and NDCG@2 (0.916).

10.5 The practical reading

Put in plain language, this specific **Basic (k=2)** report says: *the retriever is reliably good at surfacing at least one relevant document quickly for almost every question, and for most questions it surfaces everything needed – but it has one specific, real weak spot on questions that need two documents at once, where a single dominant document can crowd out a second required source entirely.* That sentence is not a guess – it is a direct, defensible reading of exactly these six numbers together, and it matches, number for number, the qualitative finding this project’s decision log recorded independently for case 4 across MMR, ParentDocumentRetriever, and RAPTOR.

10.6 Interview framing

A clean way to state this methodology out loud: *“My retrieval ground truth is annotated at the source/document level rather than the chunk level, so I evaluated retrieval quality using source-level IR metrics – Recall@K and Hit Rate@K for whether the needed evidence came back at all, Precision@K for how much of what came back was actually relevant, MRR@K and NDCG@K for how well-ranked the results were, and I added a stricter Full Coverage metric on top of all of them, because several of my test questions require synthesizing two documents at once, and no single one of the five standard metrics on its own would reliably catch a retriever that finds one of the two required sources but silently drops the other.”*